ExpenseAI - Complete Features Summary

Overview
ExpenseAI is a comprehensive expense tracking application built with React, TypeScript, and Python backend, featuring AI-powered expense categorization, receipt scanning, and financial insights.

Backend Features (Python)

Authentication & Security
- **User Registration**: Email-based user registration with secure password storage
- **User Login**: JWT-based authentication with session management
- **Password Reset**: OTP-based password reset with email support
- **Session Management**: 30-day session expiration with secure token storage
- **Password Security**: PBKDF2-HMAC-SHA256 hashing with 120,000 iterations
- **Salt Generation**: Cryptographically secure salt generation for password hashing
- **Email Notifications**: SMTP-based email delivery for password reset codes
- **Token Security**: HMAC-SHA256 token hashing for reset codes

Database Operations
- **SQLite Support**: Local SQLite database for development
- **PostgreSQL Support**: Production-ready PostgreSQL database integration
- **Database Migration**: Migration script for SQLite to PostgreSQL
- **Connection Pooling**: Efficient database connection management
- **Transaction Support**: ACID-compliant database transactions

API Endpoints
- **Authentication Endpoints**:
  - POST `/auth/register` - User registration
  - POST `/auth/login` - User login
  - GET `/auth/me` - Get current user
  - POST `/auth/logout` - User logout
  - POST `/auth/forgot-password` - Request password reset
  - POST `/auth/reset-password` - Reset password with token

- **Expense Management**:
  - GET `/expenses` - Fetch all expenses
  - POST `/expenses` - Create new expense
  - PUT `/expenses/:id` - Update expense
  - DELETE `/expenses/:id` - Delete expense

- **Income Management**:
  - GET `/income` - Fetch all income
  - POST `/income` - Create new income
  - PUT `/income/:id` - Update income
  - DELETE `/income/:id` - Delete income

Server Features
- **HTTP Server**: ThreadingHTTPServer for concurrent request handling
- **CORS Support**: Cross-origin resource sharing enabled
- **Static File Serving**: Built-in static file serving for production
- **Error Handling**: Comprehensive error handling with proper HTTP status codes
- **Request Validation**: Input validation and sanitization
- **JSON Processing**: Efficient JSON request/response handling

---

Frontend Features (React + TypeScript)

Core Components

Authentication
- **Login Component**: User login with email and password
- **Registration Component**: New user registration
- **Password Reset**: Forgot password flow with OTP verification
- **Session Persistence**: Automatic session restoration on app load

Expense Management
- **Expense Form**: Add new expenses with AI category suggestions
- **Expense List**: View and manage all expenses
- **Expense Editing**: Edit existing expense details
- **Expense Deletion**: Remove expenses with confirmation
- **AI Category Suggestions**: Automatic expense categorization
- **Category Selection**: Manual category override options

Income Management
- **Income Form**: Add new income entries
- **Income List**: View and manage all income
- **Income Editing**: Edit existing income details
- **Income Deletion**: Remove income entries

Dashboard Features

Statistics Overview
- **Total Spent**: Overall expense tracking
- **Monthly Spending**: Current month expense summary
- **AI Categorized**: AI-powered categorization statistics
- **Interactive Cards**: Clickable stat cards with detailed views
- **Detail Modals**: Detailed breakdown of expense categories

Income Overview
- **Total Income**: Overall income tracking
- **Income Breakdown**: Category-wise income distribution
- **Income Trends**: Visual income trend analysis

Charts & Visualizations
- **Spending Chart**: Visual expense distribution by category
- **Category Pie Chart**: Interactive pie chart with time range filters
- **Expense Timeline Graph**: Timeline view of expenses over time
- **Color-coded Categories**: Consistent color scheme for categories

AI Insights Panel
- **Spending Velocity**: Daily average spending calculation
- **Monthly Projections**: Predictive monthly spending estimates
- **Category Analysis**: Top spending category identification
- **Anomaly Detection**: Unusual spending pattern alerts
- **Trend Analysis**: Week-over-week spending comparisons
- **Personalized Recommendations**: AI-powered saving suggestions
- **Financial Health Score**: Overall financial health assessment

Advanced Features

AI Chatbot
- **Natural Language Processing**: Conversational AI for financial queries
- **Financial Insights**: Comprehensive financial analysis
- **Expense Categorization**: AI-powered expense classification
- **Budget Recommendations**: Personalized budget suggestions
- **Spending Analysis**: Detailed spending breakdown
- **Income Analysis**: Income source analysis
- **General Knowledge**: General Q&A capabilities

- **Math Calculations**: Built-in calculation support
- **Speech Recognition**: Voice input support
- **Speech Synthesis**: Voice output support
- **Resizable Interface**: User-resizable chat window

Receipt Scanning
- **OCR Integration**: Tesseract.js for text recognition
- **Image Preprocessing**: Enhanced image processing for better OCR
- **Receipt Parsing**: Intelligent receipt data extraction
- **Item Recognition**: Automatic item identification
- **Amount Extraction**: Automatic amount detection
- **Webcam Support**: Real-time receipt scanning via webcam
- **File Upload**: Support for image file uploads
- **Confidence Scoring**: OCR confidence assessment
- **Address Filtering**: Smart address and contact info filtering

Currency Management
- **Multi-currency Support**: Support for 8 major currencies
- **Currency Selection**: User-selectable currency preference
- **Persistent Settings**: Currency preference saved in localStorage
- **Real-time Updates**: Currency changes reflect immediately
- **Supported Currencies**:
  - Indian Rupee (₹)
  - US Dollar ($)
  - Euro (€)
  - British Pound (£)
  - Japanese Yen (¥)
  - Chinese Yuan (¥)
  - Australian Dollar (A$)
  - Canadian Dollar (C$)

UI/UX Features

Sidebar Navigation
- **Collapsible Sidebar**: Expandable/collapsible navigation
- **User Profile**: User information display
- **Quick Navigation**: Easy access to all sections
- **Settings Access**: Quick access to account settings
- **Logout Functionality**: Secure logout option

Account Settings
- **Profile Information**: User account details
- **Account Statistics**: Financial summary statistics
- **Currency Settings**: Currency preference management
- **Security Settings**: Password management
- **Dark Mode Support**: Theme toggle functionality

Responsive Design
- **Mobile-Friendly**: Responsive design for all screen sizes
- **Dark Mode**: Built-in dark theme support
- **Accessibility**: WCAG-compliant accessibility features
- **Touch Support**: Optimized for touch devices

Visual Components
- **Card Components**: Reusable card UI components
- **Button Components**: Consistent button styling
- **Input Components**: Standardized input fields
- **Modal Components**: Dialog and modal support
- **Toast Notifications**: User feedback notifications

Database Features

SQLite (Development)
- **Local Storage**: File-based database for development
- **Zero Configuration**: No setup required
- **Portable Database**: Easy database backup and migration
- **Full SQL Support**: Complete SQL feature set
- **ACID Compliance**: Transaction support and data integrity

PostgreSQL (Production)
- **Production-Ready**: Scalable database solution
- **Connection Pooling**: Efficient connection management
- **Advanced Features**: Advanced SQL features and performance
- **Data Integrity**: Robust data integrity constraints
- **Backup Support**: Comprehensive backup and recovery

Database Schema
- **Users Table**: User account information
- **Sessions Table**: Authentication session management
- **Expenses Table**: Expense records with AI suggestions
- **Income Table**: Income tracking records
- **Password Reset Table**: Password reset token management

AI & Machine Learning Features

Expense Categorization
- **Keyword Matching**: Pattern-based category detection
- **Fuzzy Matching**: Levenshtein distance for similar words
- **Confidence Scoring**: AI confidence level for suggestions
- **Category Actions**: Personalized action recommendations
- **Signal Detection**: Identifies spending patterns and signals
- **Multi-word Support**: Handles complex expense descriptions
- **8 Categories**: Pre-defined expense categories
  - Food & Dining
  - Transportation
  - Shopping
  - Entertainment
  - Bills & Utilities
  - Healthcare
  - Education
  - Travel

Financial Insights
- **Spending Velocity**: Daily average spending calculation
- **Predictive Analytics**: Monthly spending projections
- **Category Concentration**: Spending distribution analysis
- **Anomaly Detection**: Unusual spending pattern identification
- **Trend Analysis**: Week-over-week comparisons
- **Budget Recommendations**: AI-powered budget suggestions
- **Financial Health Score**: Overall financial health assessment
- **Personalized Tips**: Category-specific saving recommendations

Natural Language Processing
- **Text Analysis**: Expense description analysis
- **Tokenization**: Word and phrase extraction
- **Pattern Recognition**: Spending pattern identification
- **Context Understanding**: Context-aware categorization

Security Features

Authentication Security
- **Password Hashing**: PBKDF2-HMAC-SHA256 with 120,000 iterations
- **Salt Generation**: Cryptographically secure salt
- **Session Tokens**: JWT-based session management
- **Token Expiration**: 30-day session validity
- **Secure Storage**: LocalStorage for token persistence

API Security
- **CORS Configuration**: Controlled cross-origin access
- **Input Validation**: Request validation and sanitization
- **Error Handling**: Secure error message handling
- **Rate Limiting**: Request rate limiting (implicit)
- **SQL Injection Prevention**: Parameterized queries

Email Security
- **SMTP Authentication**: Secure email delivery
- **TLS/SSL Support**: Encrypted email transmission
- **OTP Security**: Time-limited reset codes
- **HTML Email**: Professional email templates

Development Features

Build Tools
- **Vite**: Fast build tool and dev server
- **TypeScript**: Type-safe development
- **Tailwind CSS**: Utility-first CSS framework
- **PostCSS**: CSS processing and optimization

Development Scripts
- `npm run dev` - Start Vite dev server
- `npm run backend` - Start Python backend
- `npm run dev:full` - Start full stack application
- `npm run build` - Build for production
- `npm run serve:full` - Serve production build

Code Quality
- **TypeScript**: Static type checking
- **ESLint**: Code linting (if configured)
- **Prettier**: Code formatting (if configured)
- **Component Structure**: Organized component architecture

Desktop Application Features

Electron-like Features
- **Desktop App**: Native desktop application wrapper
- **Auto-updates**: Potential for auto-update functionality
- **System Integration**: Native system integration
- **Window Management**: Custom window controls
- **Dock Integration**: macOS dock integration

App Configuration
- **Info.plist**: macOS app configuration
- **Icon Set**: Multiple icon sizes for different displays
- **Resource Bundling**: Static resource packaging

Utility Features

Helper Functions
- **Currency Formatting**: Multi-currency support
- **Date Formatting**: Consistent date display
- **ID Generation**: Unique identifier generation
- **Date Utilities**: Date manipulation helpers
- **Expense Sorting**: Date-based expense sorting

Data Processing
- **Receipt Parsing**: Intelligent receipt data extraction
- **Address Filtering**: Smart address removal
- **Item Recognition**: Automatic item identification
- **Amount Extraction: Automatic amount detection

Technology Stack

Frontend
- **React 18.3.1**: UI framework
- **TypeScript 5.6.3**: Type-safe JavaScript
- **Vite 6.0.3**: Build tool and dev server
- **Tailwind CSS 3.4.15**: CSS framework
- **Lucide React 0.468.0**: Icon library
- **Tesseract.js 6.0.0**: OCR library
- **React DOM 18.3.1**: React DOM renderer

Backend
- **Python 3.x**: Backend language
- **Standard Library**: No external dependencies required
- **psycopg2**: PostgreSQL adapter (optional)
- **smtplib**: Email sending
- **hashlib**: Cryptographic functions
- **secrets**: Secure random generation

Database
- **SQLite**: Development database
- **PostgreSQL**: Production database (optional)

Configuration

Environment Variables
- `PGHOST` - PostgreSQL host
- `PGDATABASE` - PostgreSQL database name
- `PGUSER` - PostgreSQL user
- `PGPASSWORD` - PostgreSQL password
- `PGPORT` - PostgreSQL port
- `EMAIL_ADDRESS` - SMTP email address
- `EMAIL_PASSWORD` - SMTP password
- `SMTP_SERVER` - SMTP server address
- `SMTP_PORT` - SMTP port
- `EMAIL_USE_TLS` - TLS configuration
- `EMAIL_USE_SSL` - SSL configuration
- `EMAIL_FROM_NAME` - Email sender name

Application Configuration
- Session duration: 30 days
- Password reset expiry: 1 hour

- SMTP timeout: 15 seconds
- Max JSON size: 1MB

File Structure

```
src/
├── components/
|   ├── AIChatbot.tsx          # AI-powered chatbot
|   ├── AccountSettings.tsx    # User settings
|   ├── CategoryPieChart.tsx   # Pie chart visualization
|   ├── ExpenseForm.tsx        # Expense entry form
|   ├── ExpenseList.tsx        # Expense list view
|   ├── ExpenseTimelineGraph.tsx # Timeline visualization
|   ├── IncomeForm.tsx          # Income entry form
|   ├── IncomeList.tsx          # Income list view
|   ├── IncomeOverview.tsx      # Income statistics
|   ├── InsightsPanel.tsx       # AI insights display
|   ├── ReceiptScanPanel.tsx    # Receipt scanning
|   ├── Sidebar.tsx             # Navigation sidebar
|   ├── SpendingChart.tsx       # Spending visualization
|   ├── StatsOverview.tsx       # Statistics overview
|   ├── UserProfile.tsx         # User profile component
|   ├── auth/                   # Authentication components
|   └── ui/                     # UI components
├── lib/
|   └── api.ts                  # API client
├── types/
|   └── index.ts                # TypeScript types
├── utils/
|   ├── aiEngine.ts             # AI categorization engine
|   ├── helpers.ts              # Utility functions
|   └── receiptParser.ts        # Receipt parsing
├── App.tsx                     # Main application
├── main.tsx                    # Application entry
└── index.css                   # Global styles

backend/
├── server.py                   # Python backend server
├── dev.py                      # Development server
└── migrate_to_postgres.py      # Database migration
```

Summary

ExpenseAI is a feature-rich expense tracking application that combines modern we
b technologies with AI-powered features to provide users with comprehensive fina
ncial management capabilities. The application offers:

- **Secure Authentication**: Robust user authentication and session management
- **AI-Powered Categorization**: Intelligent expense categorization with confide
nce scoring
- **Receipt Scanning**: OCR-based receipt scanning and data extraction
- **Financial Insights**: AI-generated financial insights and recommendations
- **Interactive Visualizations**: Multiple chart types for data visualization
- **Multi-currency Support**: Support for 8 major currencies
- **Responsive Design**: Mobile-friendly with dark mode support
- **Desktop Application**: Native desktop app wrapper

- **Production-Ready Backend**: Scalable Python backend with PostgreSQL support
- **Comprehensive Security**: Multiple layers of security features

The application is designed to be both user-friendly and powerful, providing users with the tools they need to track, analyze, and optimize their spending habits.